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Diesel Fuel Contaminated with Water, Gasoline, or Diesel Exhaust Fluid (DEF)

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This document contains generalized information to cover the full range of Cummins® products. For engine specific fuel system information, see the corresponding Operation and Maintenance manual and Owner's manual.

The appropriate fuel recommendations and specifications for engines can be found in the Fluids for Cummins® Products Service Manual, Bulletin 5411406. Operating the engine using fuels and additives that do **not** meet the specifications may cause symptoms including, but **not** limited to, the following:

- Crank No Start
- Runs Rough
- Low Power
- Exhaust Smoke
- Low Fuel Rail Pressure

Follow the appropriate Service Procedure depending on whether the diesel fuel system has been contaminated with water, gasoline, or DEF.

Procedure A- Diesel Fuel Contaminated with Water, Engine Started

1. Remove, drain, and clean the fuel tank to prevent reintroducing the contaminants.
2. Replace all fuel system components.
 - Metallic low pressure fuel supply lines
 - All fuel filters
 - Low pressure and high pressure fuel pumps
 - High pressure fuel lines
 - High pressure fuel rail
 - All fuel injectors and high pressure connectors
 - Metallic low pressure return lines
 - All fuel inlet and return valves.
3. Flush any remaining fuel lines with clean diesel fuel.
4. Install and fill the tank with recommended fuel. See Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Procedure B- Diesel Fuel Contaminated with Water, Engine not Started

Follow Procedure B **only** if the ignition key has **not** been inserted or turned to the ON position. Turning the key to the RUN position may operate the fuel priming pump and supply the contaminated fuel to the high-pressure fuel system.

1. Verify there is **not** water in the Fuel Water Separator.
 - a. If water is present, reference procedure A.
2. Remove, drain, and clean the fuel tank.
3. Install and fill the tank with recommended fuel. See Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Procedure C - Diesel Fuel Contaminated with Gasoline, Engine not Started

Follow Procedure C **only** if the ignition key has **not** been inserted or turned to the ON position. Turning the key to the RUN position may operate the fuel priming pump and supply the contaminated fuel to the high-pressure fuel system.

1. Remove, drain, and clean the fuel tank.
2. Install and fill the tank with recommended fuel. See Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Procedure D - Diesel Fuel Contaminated with Gasoline, Engine Started

1. Remove, drain, and clean the fuel tank to prevent reintroducing the contaminants.
2. Perform Fuel System Flush.
 - a. Connect external fuel hoses to the fuel supply and return lines from the fuel tank.
 - b. Place the supply hose into a suitable container containing recommended diesel fuel.
 - c. Place the return hose end into a suitable empty container.
 - d. Flush the fuel system with the diesel fuel from the first container by cycling the ignition key to operate the electric lift pump. If an electric lift pump is **not** available on the engine, external cranking may be required to flush the fuel system.
 - e. Dispose of flushed fuel in accordance with local regulations. Do **not** re-use flushed fuel.
3. Replace all fuel filters on the vehicle.
4. Install and fill the tank with recommended fuel. See Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Procedure E - Diesel Fuel Contaminated with DEF, Engine Not Started

Follow Procedure E **only** if the ignition key has **not** been inserted or turned to the ON position. Turning the key to the RUN position may operate the fuel priming pump and supply the contaminated fuel to the high-pressure fuel system.

1. Remove, drain, and clean the fuel tank.
2. Install and fill the tank with recommended fuel. See Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Procedure F - Diesel Fuel Contaminated with DEF, Engine Started

1. Remove, drain, and clean the fuel tank to prevent reintroducing the contaminants.
2. Replace all fuel system components.
 - Low pressure fuel supply lines
 - All fuel filters
 - Low pressure and high pressure fuel pumps

- High pressure fuel lines
- High pressure fuel rail
- All fuel injectors and high pressure connectors
- Low pressure return lines
- All fuel inlet and return valves.

3. Install and fill the tank with recommended fuel. See Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Document History

Date	Details
2020-8-4	Module Created
2020-9-14	Added diesel fuel contamination with water information, Procedure A, and Procedure B. Adjusted other procedures accordingly.
2020-10-2	Added engines.
2022-2-14	Added engines.

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